Week 8 Software Lesson: Proportional Hazards Regression in R

Goals

The goal of this software lesson is to learn how to

- fit a proportional hazards regression model and obtain estimates for the effect of a predictor in terms of log hazards and in terms of hazard ratios, and
- obtain model diagnostics to check assumptions and evaluate the model fit.

Required Packages

You will need to install and load the following packages in R for this software lesson:

- `survival`
- `ggfortify`
- `survminer`
- `multcomp`

DATASET

Let's use the Laryngoscope dataset again.

Recall that the dataset, data dictionary, and dataset introduction can be found on Canvas, titled *Laryngoscope.csv*, *Laryngoscope Data Dictionary.pdf*, and *Laryngoscope Dataset Introduction.pdf*, respectively.

```
lar <- read.csv(file = file.choose(), header = TRUE)
```

In a previous lesson, you examined successful first attempt intubations. This week, you will compare time-to-intubation in first attempt between each laryngoscope. The models we will explore are:

- **Model 1:** Is there a difference in the time-to-intubate on the first try for the different types of laryngoscopes?
- **Model 2:** Is there a difference in the time-to-intubate on the first try for the different types of laryngoscopes, controlling for BMI and ease of intubation?

EXPLORATORY DATA ANALYSIS (EDA)

As mentioned in a previous lesson, when you have an outcome that is a time-to-event variable, you use a Kaplan-Meier graph to best visualize time-to-event data. (Note: for completeness, the instructions from the previous lesson will be presented again.)

To create a Kaplan-Meier survival graph, we need to first calculate the estimated survival via the Kaplan-Meier method by using the `survfit()` function from the `{survival}` package. Unlike before where we can use a single variable as the response variable, time-to-event data need to take into account two variables: one for the time variable and one for the event indicator variable. To create this single time-to-event variable (our response variable), we use the `Surv()` function and specify the time variable and then the event indicator variable. In this case, *attempt1_time* is the time variable (or the time index), and *attempt1_S_F* is the event indicator.

To complete the model, this new time-to-event variable (with censoring built in) should be followed by ~ and then a `1` if you want overall survival (not by group) or the grouping variable (e.g., *Randomization*).

Note that we will must create the model before we can create a Kaplan-Meier curve.

```
kmsurv<-survfit(Surv(attempt1_time, attempt1_S_F) ~ Randomization, data = lar)
```

To plot the Kaplan-Meier curves, use the `autoplot()` function from the `{ggfortify}` package. There are additional plotting options you can use to customize your plot (e.g., `xlab()` for specifying the x-axis label, `ylab()` for specifying the y-axis label, add a title, subtitle, and caption to the plot with `labs()`). These are added by using a `+` sign in between each function after the `autoplot()` function. (Note: These additional plotting options can continue over many lines.)

```
autoplot(kmsurv) +
  xlab("Time (in seconds)") +
  ylab("Percent Not Succesfully Intubated") +
  labs(title = "Time to Intubation", subtitle = "By Laryngoscope Type",
       caption = "First Attempt Only") +
  theme_bw()+ #optional
  theme(plot.title = element_text(hjust=0.5, face="bold"),
        plot.subtitle = element_text(hjust=0.5))
```

The output includes one curve per group, including 95% confidence regions for the estimated survival probabilities. Each + on the plot indicates censored data (i.e., observations that did not have the event, in this case, a successful intubation).

Note that the curves do not cross, which may indicate the hazards are proportional (an assumption for Cox proportional hazards regression).

MODEL FITTING

Model 1: Cox proportional hazards regression with a categorical predictor

To fit a proportional hazards regression model, use the `coxph()` function from the `{survival}` package. (Note: The structure looks similar to what you saw in previous lessons.) The arguments for the function are the model formula (e.g., `Surv(attempt1_time, attempt1_S_F) ~ Randomization`), followed by the name of the data set (e.g., `data = lar`).

```
model1 <- coxph(Surv(attempt1_time, attempt1_S_F) ~ Randomization, data = lar)

summary(model1)
```

The output contains three sections. The first section (row starting after `n= #, number of events= #`) provides the regression coefficient estimates in terms of log hazards (`coef`) and hazard ratios (`exp(coef)`), the standard error for the log hazards (`se(coef)`), the *z*-statistic for log hazard regression coefficient (`z`, which is the ratio of each regression coefficient to its standard error), and its *p*-value (`Pr(>|z|)`).

The second section (row starting after `Signif. codes:`) provides the hazard ratio (`exp(coef)`), the inverse hazard ratio (`exp(-coef)`), and the 95% confidence interval for the hazard ratio (`lower .95` and `upper .95`).

The third section provides three tests for overall significance of the model: The `Likelihood ratio test` (LRT), `Wald test`, and `Score (logrank) test`. These all have the omnibus null hypothesis that all of the

β's are zero. For samples with large n, they will give similar results. If the sample size is small, they may differ somewhat. The LRT is generally the preferred test, since it behaves better for small sample sizes.

Checking the proportional hazards assumption

Recall that one of the approaches to check the proportional hazards assumption is to calculate the Schoenfeld residuals and make a plot of the residuals verses time.

To calculate Schoenfeld residuals, first, use the `cox.zph()` function on the proportion hazards regression model from the `{survival}` package. Then, generate plots of the Schoenfeld residuals with the `ggcoxzph()` function in the `{survminer}` package, specifying the `cox.zph` object.

```
model1_resid <- cox.zph(model1)

ggcoxzph(model1_resid)
```

Only a single plot of Schoenfeld residuals verses time will be generated because there's only one predictor. To evaluate the proportional hazards assumptions using the plot, determine whether there appears to be time trends. The proportional hazards assumption is met if there is a lack of a trend over time.

This plot also features the result of the proportionality test (it will run automatically and will present the p-value result from the test). The null hypothesis for the test is that there is no time-dependent trend in the residuals.

The p-value listed at the top of the plot is testing the proportional hazards assumption for all the listed predictors at once. Since there is only one predictor in this model, this is the same p-value as listed on the graph for the `Randomization` predictor.

NOTE: SAS and R use different algorithms to perform a proportionality test, so the results will differ. Most of the time the differences will be minimal and the conclusions will be the same, but sometimes you will end up at a different conclusion.

Model 2: Multiple Cox proportional hazards regression

In this model, we will control for *BMI* and ease of intubation (*Mallampati*). You need to make a new variable that turns the *Mallampati* into a factor type variable (in order for the pairwise comparisons code to work in the next part).

```
lar$m.f <- factor(lar$Mallampati)

model2 <- coxph(Surv(attempt1_time, attempt1_S_F) ~ Randomization + BMI + m.f, data = lar)

summary(model2)
```

Pairwise comparisons for multi-categorical predictors

The ease of intubation variable (*Mallampati*) has 4 categories, but we only see 3 of the categories in comparison to the reference group of 1 (which represents "Full Visibility") in the model output.

To conduct pairwise comparisons for all categories of a predictor that has multiple categories, use the `glht()` function from the `{multcomp}` package. The arguments for the `glht()` function (stands for General Linear Hypotheses) are `model=`, to specify the Cox regression model, and `linfct=mcp()`, to specify the variable that you want to make pairwise comparisons and the multiple comparison adjustment method (`"Tukey"`). To obtain the hazard ratios comparing pairwise categories and their confidence intervals, use the `confint()`

function on the `glht` object (e.g., `cox.multcomp`), which gives us the confidence interval for the log hazards, and then exponentiate.

```
cox.multcomp <- glht(model2, linfct = mcp(m.f = "Tukey"))

###Obtain HRs and their CIs
exp(confint(cox.multcomp)$confint)
```

The first column denotes the two categories that are being compared. The second column (`Estimate`) provides the hazard ratios between the two groups and the subsequent columns (`lwr` and `upr`) provides the lower and upper limits of the confidence interval (which took into account multiple comparisons). You determine statistical significance between the groups as you would for any other hazard ratio (i.e., look to see if 1 is in the interval or not).

Checking the proportional hazards assumption

We should also check the proportional hazards assumption for Model 2. The code is similar to the code presented earlier in Model 1.

```
model2_resid <- cox.zph(model2)

ggcoxzph(model2_resid)
```

One plot will be created for each explanatory variable in the model, however, focus only on the one (or two) plot(s) for the variable(s) of most interest when checking the proportional hazards assumption. In this case, since the second research question is focused on the interventions, the plot of Schoenfeld residuals vs. time for the `Randomization` predictor is the one we should examine.

Additional Resources

For additional R resources on proportional hazards regression, see:

- Cox Proportional-Hazards Regression for Survival Data in R (from John Fox and Sanford Weisberg)
- Cox Proportional Hazards Model (from STDHA)
- Cox Model Assumptions (from STDHA)